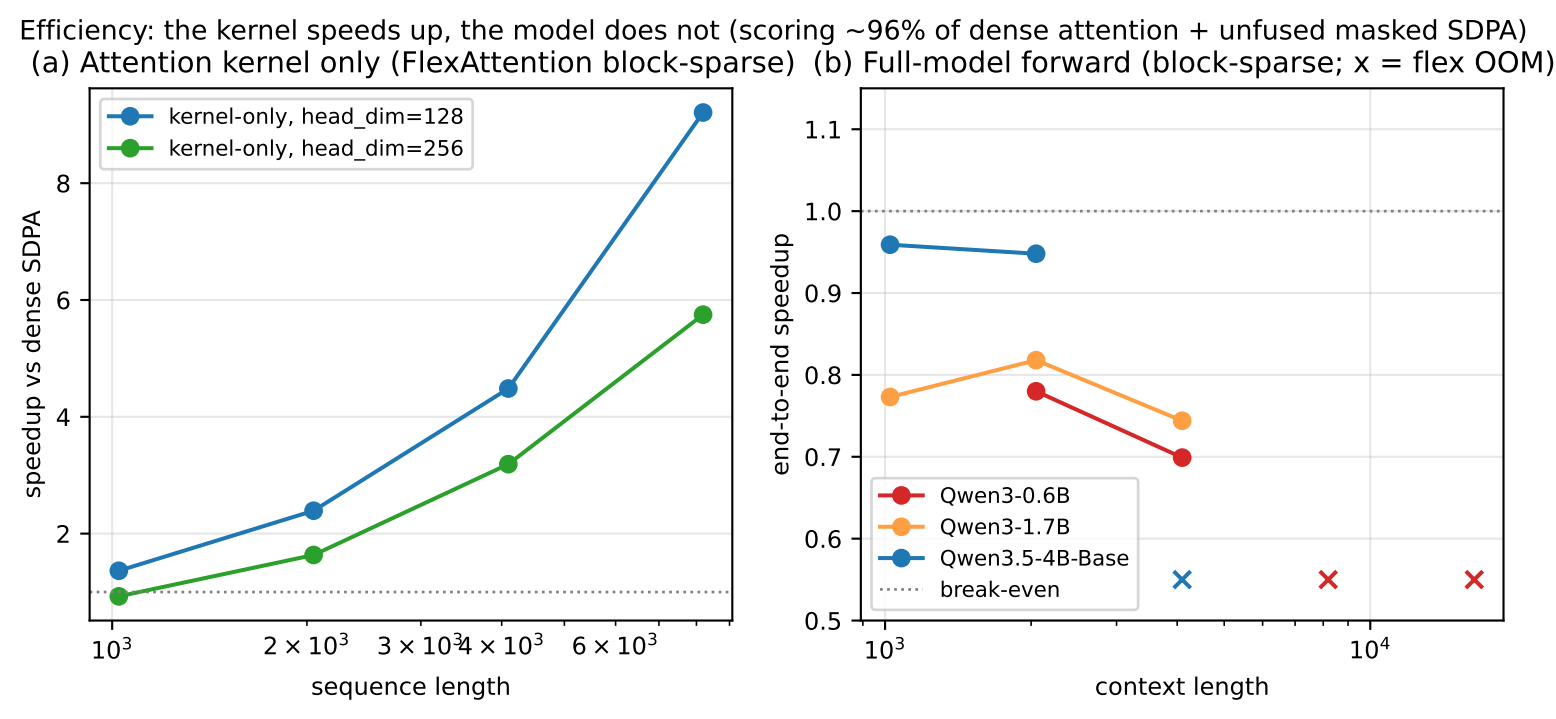
Efficiency: the kernel speeds up, the model does not (scoring ~96% of dense attention + unfused masked SDPA)
(a) Attention kernel only (FlexAttention block-sparse)
(b) Full-model forward (block-sparse; x = flex OOM)
kernel-only, head_dim=128
kernel-only, head_dim=256
speedup vs dense SDPA
sequence length
end-to-end speedup
context length
Qwen3-0.6B
Qwen3-1.7B
Qwen3.5-4B-Base
break-even